

Ramdeobaba University, Nagpur

IPR CELL

DESIGN REQUISITES DRAFT

1. Full Name and Address of the inventor(s) and applicant(s)

SN	Name of applicant/Inventor	Designation Faculty /UG/PG student/Research scholar	E mail ID	Mobile Number	Address
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2. What is the suggested title of the Design?

THIRD EYE FOR THE BLIND - Advanced Wearable Technology Enhancing Mobility for the Visually Impaired

3. Write one page information about your design. Like working, Advantages, innovations, new etc.

The "**Third Eye for the Blind**" is an innovative wearable technology designed to enhance mobility for visually impaired individuals by providing real-time obstacle detection through an intuitive auditory feedback system. The device is seamlessly integrated into a pair of glasses, making it highly discreet and user-friendly. The goal of this design is to empower visually impaired users to navigate their surroundings safely, independently, and with greater confidence.

How the Design Works

The primary working mechanism of the "Third Eye for the Blind" is based on the use of **ultrasonic sensors** that continuously scan the environment for obstacles. These sensors measure the distance to nearby objects by emitting sound waves and calculating the time taken for the echo to return. If an object is detected within a predefined range (such as 100 cm), the **Arduino Nano** microcontroller processes this data and activates a **buzzer**. The buzzer generates a sound at 85 dB, alerting the user to the presence of an obstacle. The sound serves as a warning, enabling the user to take necessary action to avoid the obstacle.

The device is powered by a **9V battery**, and the system is controlled by a single **push button** that allows the user to activate or deactivate the device. The components, including the ultrasonic sensor, buzzer, Arduino, and battery, are carefully mounted onto a pair of glasses to ensure ergonomic design and comfort for long-term wear.

Advantages of the Design

1. **Enhanced Mobility:** By providing real-time auditory alerts, the device allows users to navigate their surroundings without relying on physical aids like canes or guide dogs. This enables greater freedom and independence.
2. **Cost-Effective:** The use of readily available components such as the Arduino Nano, ultrasonic sensors, and a buzzer ensures that the device is both affordable and easy to manufacture on a large scale. This affordability makes it accessible to a wider population, including those in low-income regions.
3. **Intuitive Feedback:** The sound-based feedback mechanism is simple and immediate, eliminating the need for the user to learn complex interfaces or interpret complicated signals. The use of a buzzer ensures that the alerts are easily distinguishable and provide clear guidance on obstacle proximity.
4. **Compact and Wearable:** The integration of the components into a pair of glasses offers a practical and unobtrusive solution. The glasses are lightweight and designed for comfort, ensuring that the user can wear them throughout the day without discomfort.

Innovative Features

- **Auditory Alerts:** Unlike most wearable assistive devices, which use haptic (vibration) feedback, the "Third Eye for the Blind" employs auditory feedback through a buzzer. This choice enhances the device's usability, as sound is more easily perceived by individuals who may not be accustomed to vibratory cues.
- **Simple Design:** The system operates with just one push button for activation and deactivation, making it extremely simple for users of all ages and technical skill levels to operate. The simplicity of the design also means fewer components, reducing the risk of malfunction.
- **Energy Efficiency:** The design prioritizes low power consumption to extend the battery life of the device, ensuring that users can rely on it for several hours without needing frequent recharges or battery replacements.

New and Novel Aspects

One of the most significant innovations in this design is its simplicity and accessibility. While there are other obstacle detection systems available for visually impaired users, many of these devices are either too complex or expensive to be widely adopted. The "Third Eye for the Blind" leverages basic, off-the-shelf components like ultrasonic sensors and buzzers to provide a low-cost solution without compromising on effectiveness. Additionally, the use of glasses as the mounting platform is novel; it keeps the device discreet and convenient, integrating seamlessly into the user's daily routine.

The adaptability of the device is another novel feature. The threshold distance for detecting obstacles can be adjusted based on user preferences or the environment they are navigating. For instance, in crowded urban settings, the sensitivity can be fine-tuned to detect closer objects, while in more open spaces, it can be extended for long-range detection.

Future Potential and Scalability

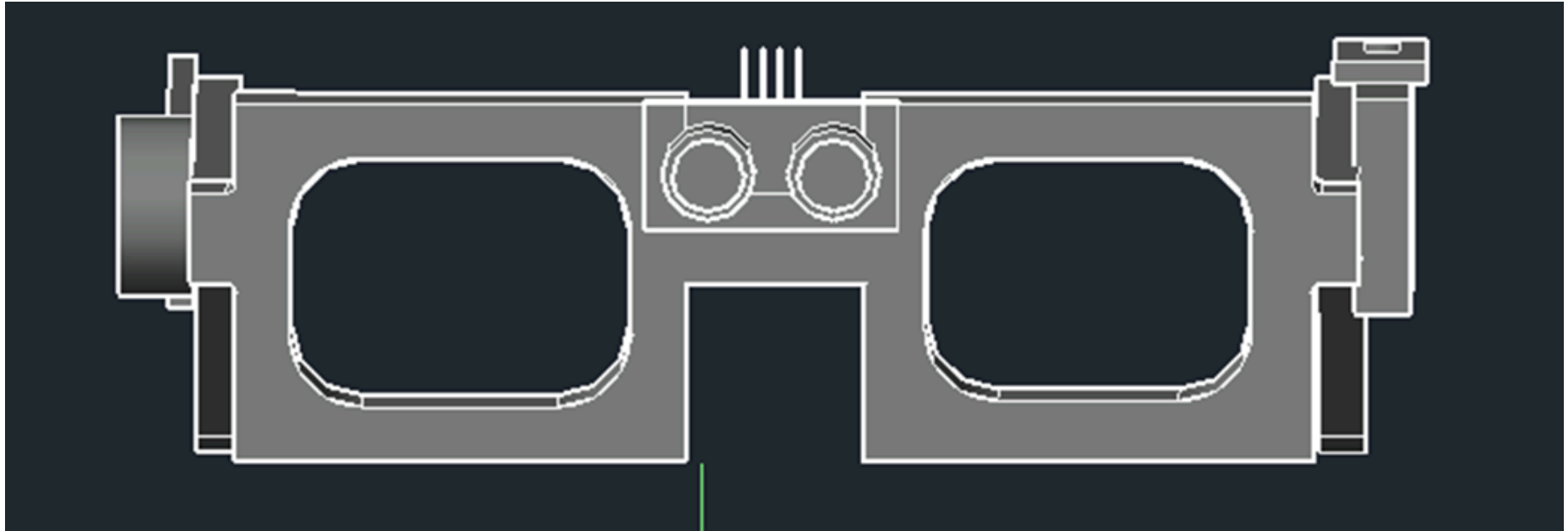
The design holds great potential for future enhancements and scalability. For instance, by incorporating **machine learning algorithms**, the system could be trained to differentiate between various types of obstacles, providing more specific feedback to the user (e.g., a distinct sound for detecting cars versus pedestrians). Additionally, future iterations could include **wireless connectivity** to integrate the device with smartphones or other wearables, allowing for real-time data sharing and enhanced user control.

The "Third Eye for the Blind" can be produced on a mass scale due to its low-cost components and straightforward assembly process. This opens up opportunities for partnerships with NGOs and healthcare organizations to distribute the device to visually impaired communities around the world. The ultimate aim is to create a product that not only meets the immediate needs of visually impaired users but also contributes to a broader societal shift toward inclusive and accessible technology.

In conclusion, the "Third Eye for the Blind" offers an innovative, simple, and cost-effective solution to a critical issue faced by visually impaired individuals. By focusing on usability, affordability, and intuitive design, it stands to make a significant impact on the quality of life for millions of people worldwide.

4. Photographs/Drawings of the article in **SEVEN VIEWS** [front, rear, top plan, bottom plan, left and right side elevation and isometric] .These seven views/ design should be a line diagram, photographs or CAD drawings

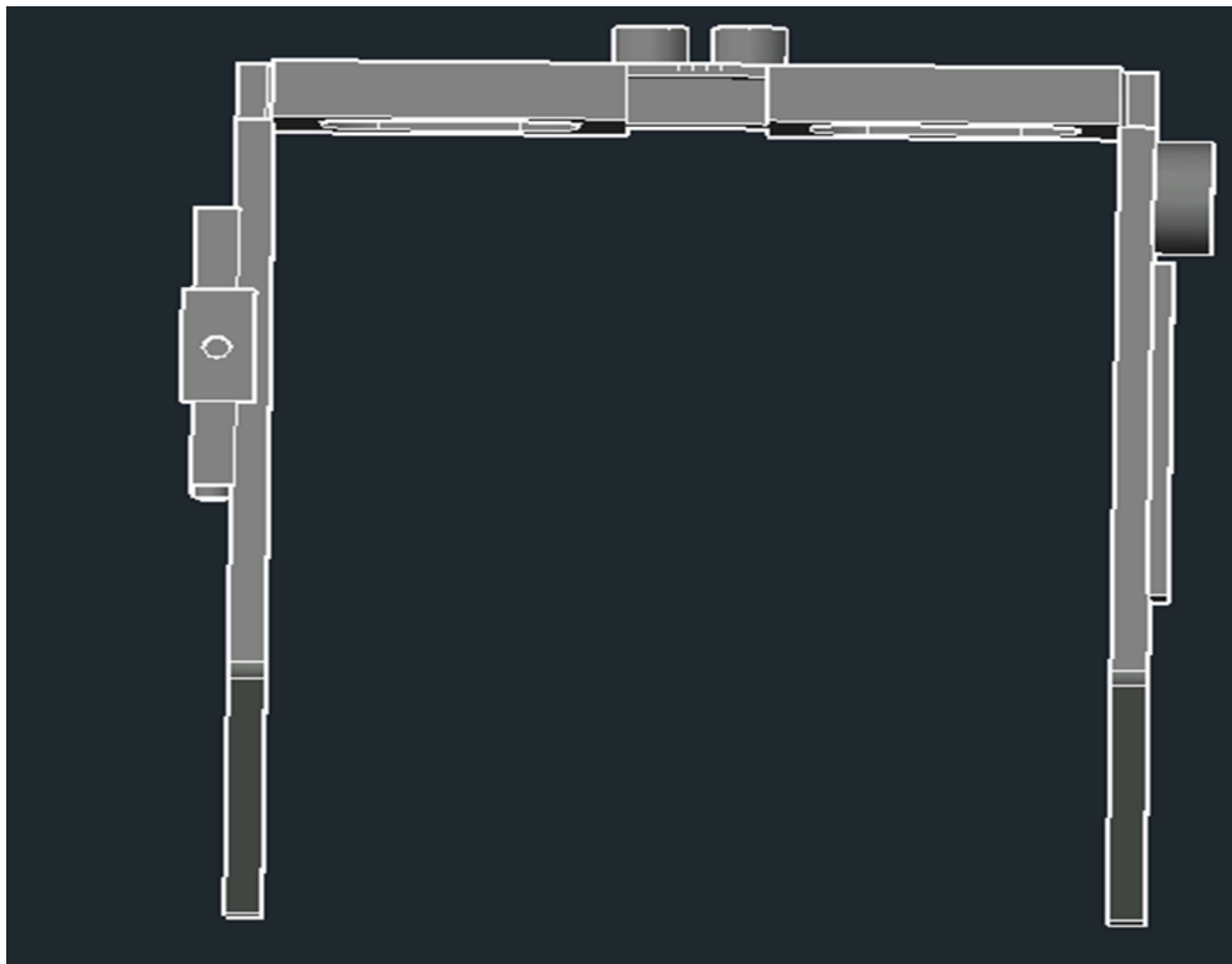
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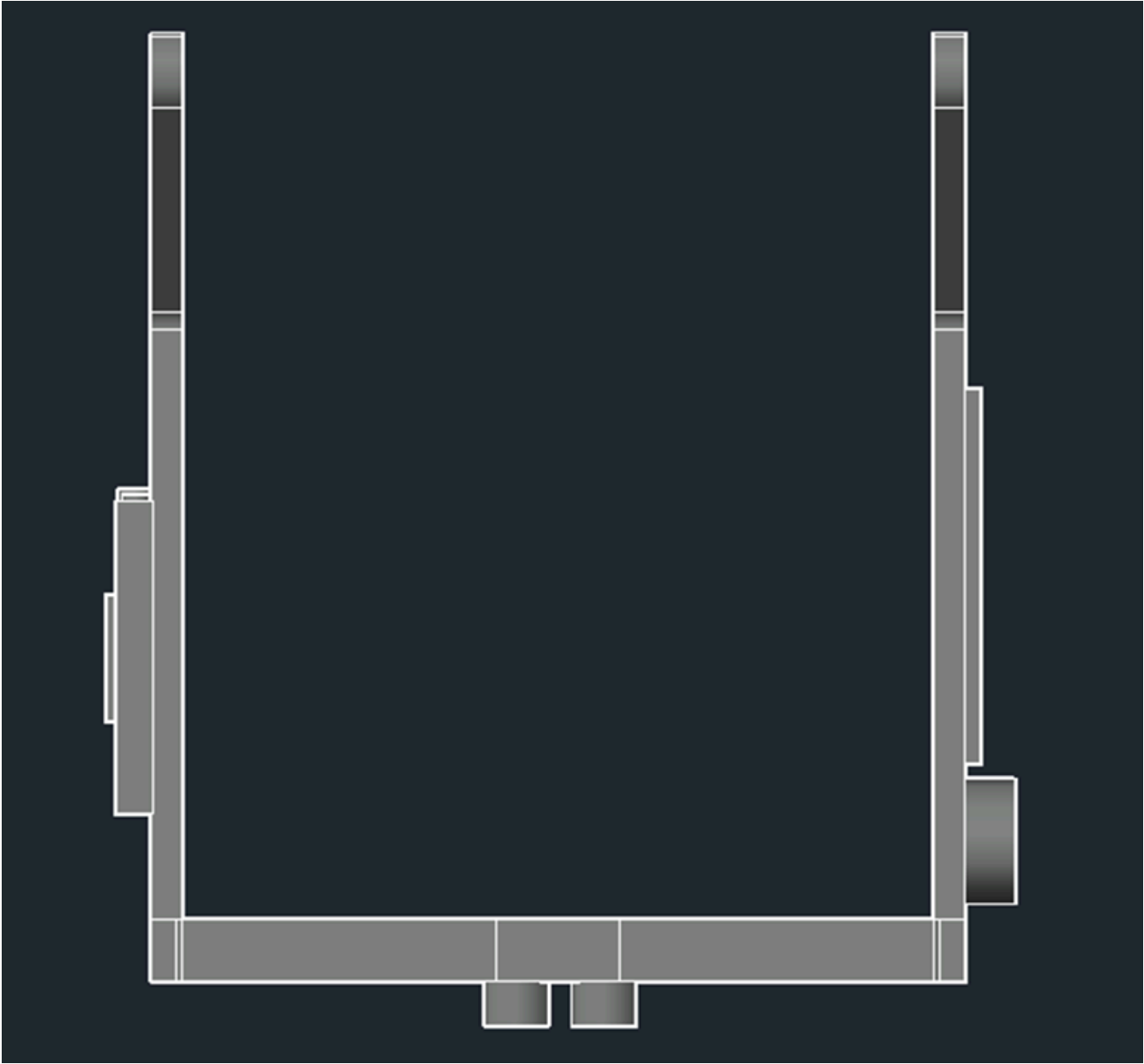
Rear View:



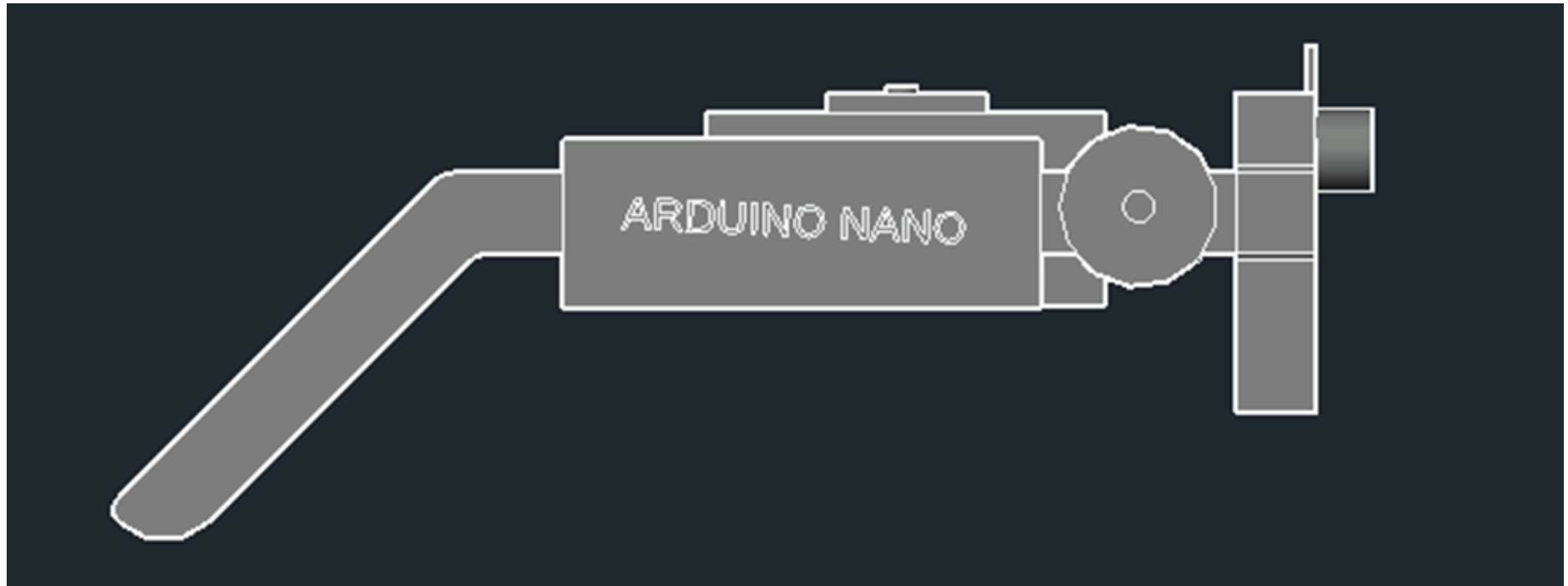
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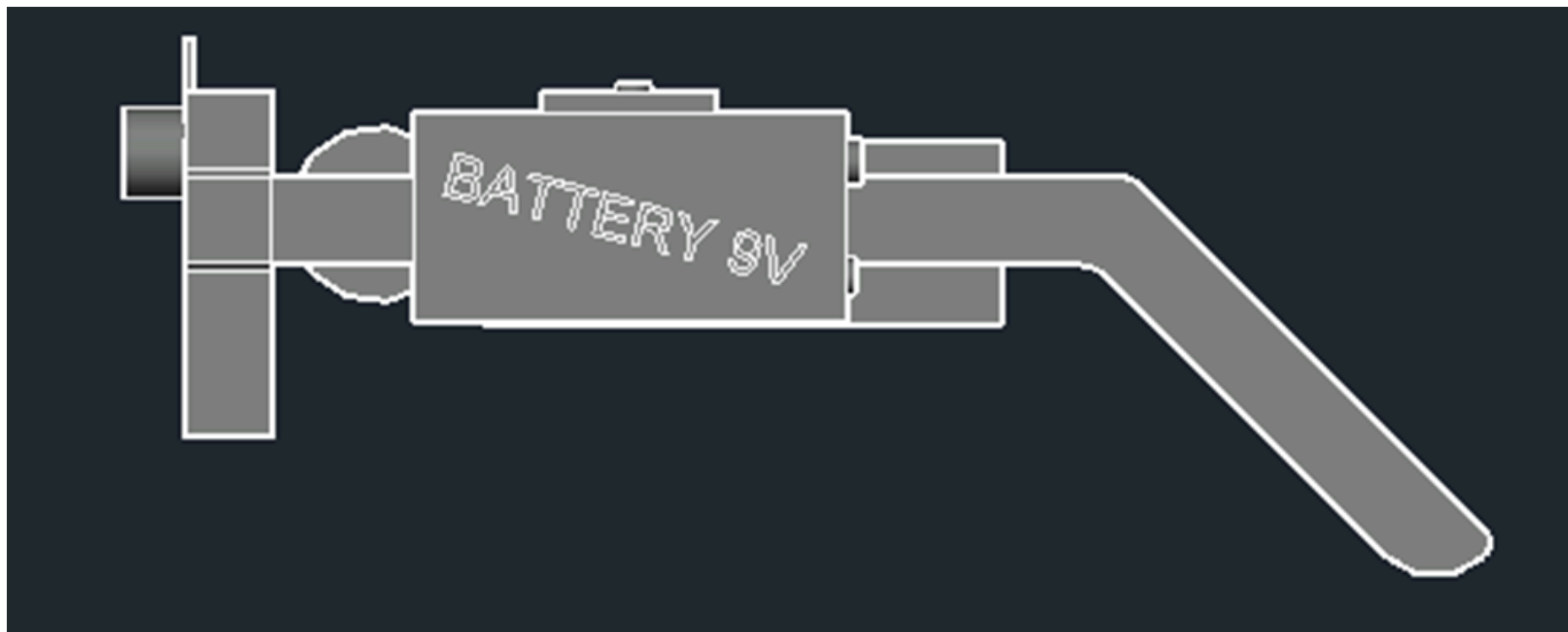
Bottom View:



Left Side View:



Right Side View:



Isometric View:

